PORTFOLIO PROJECT · DATA & AI

Change a marketing strategy in one sentence

NBO Segment Editor: a governed Next Best Offer engine for prepaid telecom, driven in plain English with RAG.

[Your name] · Inspired by a real NBO system built for a telecom operator. All data and the segmentation grid shown here are synthetic.

THE PROBLEM

One offer per subscriber, but the rules are hard-coded

1

Slow

Every rule change needs a developer and a production release.

2

Risky

One wrong bound can leave customers with no offer, or with two.

3

Blind

Nobody sees the impact before it goes live: who moves, and where?

THE PROMISE

The business owns the rules. The AI proposes. The engine verifies and applies.

Take the developer out of the loop, without giving up control.

ARCHITECTURE

Three levels, plus a trace of everything

LEVEL 3	**RAG editor**	Turns a sentence into a structured change (JSON), or declines.
LEVEL 2	**Automatic control**	Validates, simulates the impact on the base, applies or rejects with the reason.
LEVEL 1	**NBO engine**	Assigns every subscriber to exactly one segment, hence one offer.
ACROSS	**Traceability**	Every request, applied or rejected, is logged with a unique id.

The AI only acts at level 3: everything that touches the data is deterministic.

The strategy is a table, not code

How the engine decides

1. Compute 6 behavioural variables per subscriber
2. Compare the subscriber with every active rule
3. Keep the rules whose conditions are all true
4. Exactly 1 rule → its segment and its offer
5. 0 rules → UNASSIGNED: no offer
6. 2+ rules → MULTIPLE: conflicting offers

One rule = one row

```
tenure ∈ [min, max)
AND renewals ∈ [min, max)
AND days late ∈ [min, max)
AND roaming compatible
AND risk ∈ allowed combinations
```

→ 1 segment → 1 offer intensity

Changing the strategy = changing a value in a row. No code, no release.

Six behavioural variables describe every subscriber

Variable	What it measures	Bands used in the grid
Tenure	Days since the line was activated	< 60 Welcome · 60–149 Onboarding · 150+ mature
Renewals (6 months)	Main-bundle renewals over 180 days	6+ Champion · 4–5 Regular · 1–3 Occasional · 0 Silent
Days late	Days past the validity of the last bundle	< 4 Active · 4–14 Slipping · 15–45 Lapsed · 46+ Lost
Spend trend	Spending going up, flat or down	POS · FLAT · NEG
Churn risk	Churn-model probability, banded	LOW · MED · HIGH
Roaming	Recent roaming activity	Y / N: a late customer may just be travelling

Spend trend and churn risk combine into one risk code, e.g. POS-MED or NEG-HIGH. All bands are fictional (portfolio grid).

Loyalty × lifecycle stage: one cell per customer

	Active < 4 days	Slipping 4–14	Lapsed 15–45	Lost 46+
New customers tenure < 150 days	**C01 Welcome Journey** NONE · < 60 days		**C02 Onboarding Boost** NORMAL · 60–149 days	
Champion 6+ renewals	**C03 Upsell** NORMAL	**C04 Protect** AGGRESSIVE	**C05 Champion Nudge** NORMAL	**C06 Champion Recovery** AGGRESSIVE
Regular 4–5 renewals	**C07 Regular Grow** NORMAL	**C08 Regular Nudge** AGGRESSIVE	**C09 Win-back** AGGRESSIVE · not roaming	**C10 Traveller** NONE · roaming
Occasional 1–3 renewals	**C11 Occasional Activate** NORMAL		**C12 Occasional Win-back** VERY_AGGRESSIVE	
Silent 0 renewals	**C13 Silent Reactivate** AGGRESSIVE		**C14 Silent Last Call** VERY_AGGRESSIVE	

C03 / C04: active Champions split by risk (healthy → upsell, at risk → protect). C09 / C10: lapsed Regulars split by roaming. Fictional grid: no gaps, no overlaps.

How SUB000001 gets their Next Best Offer

SUB000001

Tenure: 3,330 days
Renewals (6m): 7
Days late: −2 (ahead)
Roaming: N
Risk code: POS-MED

Result

C03 Champion Upsell → NORMAL offer

1. **3,330 days ≥ 150**
   mature customer: not Welcome, not Onboarding
2. **7 renewals ≥ 6**
   family Champion
3. **−2 days late < 4**
   stage Active: candidates C03 and C04
4. **POS-MED is a healthy code**
   in C03's list, not in C04's → exactly 1 rule

Another one: SUB000006 (253 days, 5 renewals, 4 days late, NEG-HIGH) → C08 Regular Nudge → AGGRESSIVE.

The rule grid, as the app shows it

Segments & data — how the NBO engine assigns each customer

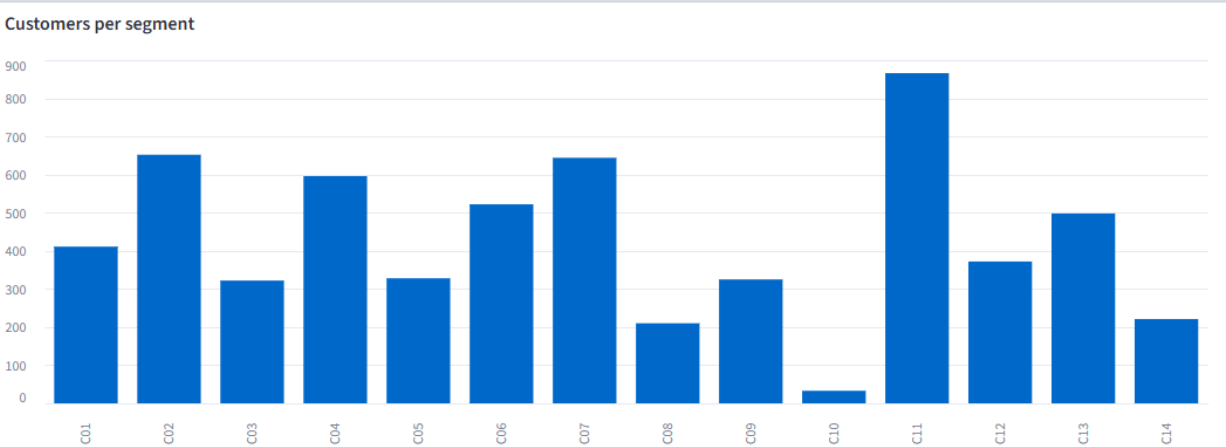
Subscribers	Active segments	Without a segment	In two segments
6,000	14	0	0

Rule grid — one row per segment (min inclusive, max exclusive)

segment_id	segment_name	renewal_group	lapse_group	tenure_min	tenure_max	renew_min	renew_max	late_days_min	late_days_max	roaming_rule	risk_combos	offer_intensity	is_active
C01	Welcome Journey	WELCOME	ANY	0	60	0	99	-999	999	ANY	ANY	NONE	Y
C02	Onboarding Boost	ONBOARDING	ANY	60	150	0	99	-999	999	ANY	ANY	NORMAL	Y
C03	Champion Upsell	CHAMPION	ACTIVE	150	99999	6	99	-999	4	ANY	|POS-LOW|POS-ME	NORMAL	Y
C04	Champion Protect	CHAMPION	ACTIVE	150	99999	6	99	-999	4	ANY	|POS-HIGH|FLAT-M	AGGRESSIVE	Y
C05	Champion Nudge	CHAMPION	SLIPPING	150	99999	6	99	4	15	ANY	ANY	NORMAL	Y
C06	Champion Recovery	CHAMPION	LAPSED_OR_LOST	150	99999	6	99	15	999	ANY	ANY	AGGRESSIVE	Y
C07	Regular Grow	REGULAR	ACTIVE	150	99999	4	6	-999	4	ANY	ANY	NORMAL	Y
C08	Regular Nudge	REGULAR	SLIPPING	150	99999	4	6	4	15	ANY	ANY	AGGRESSIVE	Y
C09	Regular Win-back	REGULAR	LAPSED_OR_LOST	150	99999	4	6	15	999	N	ANY	AGGRESSIVE	Y
C10	Regular Traveller	REGULAR	LAPSED_OR_LOST	150	99999	4	6	15	999	Y	ANY	NONE	Y

"Segments & data" tab: 6,000 subscribers · 14 active segments · 0 without a segment · 0 in two segments.

Where the 6,000 customers are



segment_id	segment_name	offer_intensity	customers	share
C01	Welcome Journey	NONE	411	6.8 %
C02	Onboarding Boost	NORMAL	652	10.9 %
C03	Champion Upsell	NORMAL	322	5.4 %
C04	Champion Protect	AGGRESSIVE	596	9.9 %
C05	Champion Nudge	NORMAL	328	5.5 %
C06	Champion Recovery	AGGRESSIVE	522	8.7 %
C07	Regular Grow	NORMAL	644	10.7 %
C08	Regular Nudge	AGGRESSIVE	210	3.5 %
C09	Regular Win-back	AGGRESSIVE	325	5.4 %
C10	Regular Traveller	NONE	33	0.5 %

Customers per offer intensity

444

NONE · 7.4%

2,812

NORMAL · 46.9%

2,151

AGGRESSIVE · 35.9%

593

VERY_AGGRESSIVE · 9.9%

Largest segment: C11 Occasional Activate, 866 customers (14.4%).

Every subscriber → one segment → one offer

Subscriber	Tenure	Renewals	Days late	Risk	Segment	Offer
SUB000000	124	4	7	NEG-HIGH	C02 Onboarding Boost	NORMAL
SUB000001	3,330	7	−2	POS-MED	C03 Champion Upsell	NORMAL
SUB000002	57	1	−10	FLAT-MED	C01 Welcome Journey	NONE
SUB000003	2,253	5	−9	NEG-HIGH	C07 Regular Grow	NORMAL
SUB000004	1,415	0	−7	NEG-MED	C13 Silent Reactivate	AGGRESSIVE
SUB000005	1,892	4	1	FLAT-HIGH	C07 Regular Grow	NORMAL
SUB000006	253	5	4	NEG-HIGH	C08 Regular Nudge	AGGRESSIVE
SUB000008	2,893	9	2	FLAT-LOW	C03 Champion Upsell	NORMAL

Extract of data/subscribers.csv with the segment computed by the engine. 100% synthetic data.

LEVEL 2 · AUTOMATIC CONTROL

Five barriers between a sentence and the data

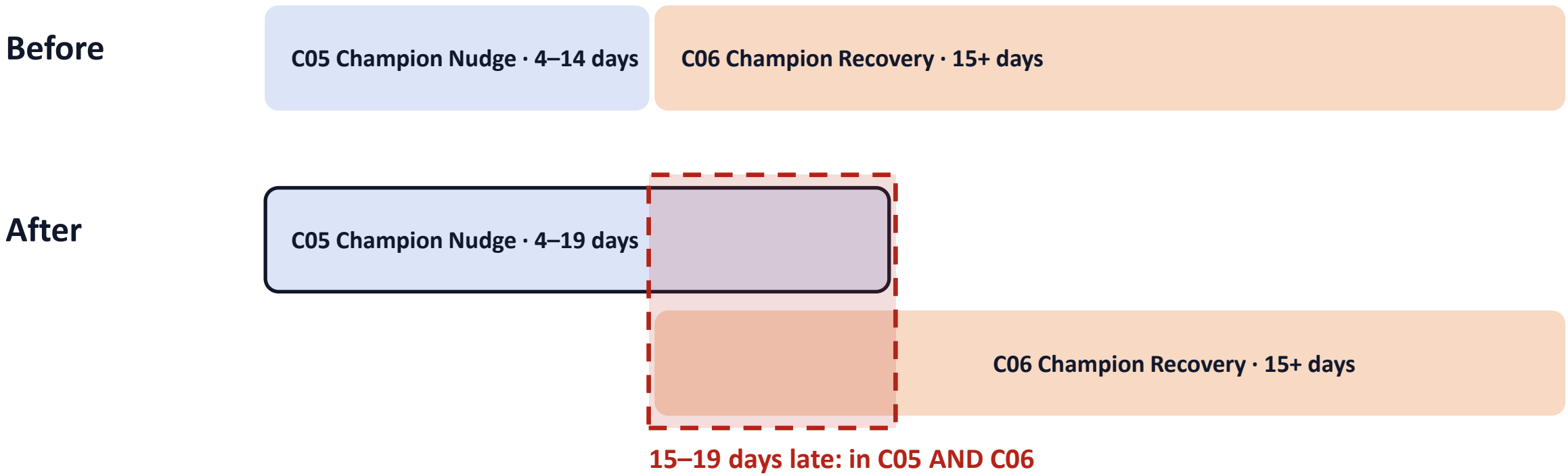
1	**The AI can decline**	Unknown segment, new variable, out of scope	`REJECTED_NO_CHANGE`
2	**Format validation**	7 checks: field, values, bounds, empty change…	`REJECTED_VALIDATION`
3	**Impact simulation**	All 6,000 customers recomputed before / after	`figures, per segment`
4	**Impact check**	Two segments → rejected · no segment → warning	`REJECTED_IMPACT`
5	**Traced application**	Rules copied before / after + a log line	`APPLIED`

No human step: if everything passes it is applied; otherwise it is rejected with the reason.

Ten guardrails, ten real tests

Control	Change tested	Message returned by the engine
Unknown segment	disable C99	Unknown segment_id 'C99'.
New variable	add "age"	Field 'age' is not editable (adding a variable is not allowed).
Value not allowed	intensity "SUPER"	offer_intensity must be one of [NONE, NORMAL, …].
Risk format	risk "HIGH"	risk_combos must be ANY or like |POS-LOW|FLAT-MED|.
Non-integer bound	tenure_max = "ninety"	tenure_max must be an integer, got 'ninety'.
Inconsistent bounds	C05 days-late min = 40	Change would make a MIN >= MAX bound.
Empty change	C07 NORMAL → NORMAL	No-op: C07 offer_intensity is already 'NORMAL'.
Nothing applicable	"Premium Gold" segment	No applicable change: … not in the current rules.
Overlap	C05 days-late max 15 → 20	**REJECTED: 45 customers would match two segments.**
No segment	disable C14	**APPLIED + warning: 221 customers will no longer receive any offer.**

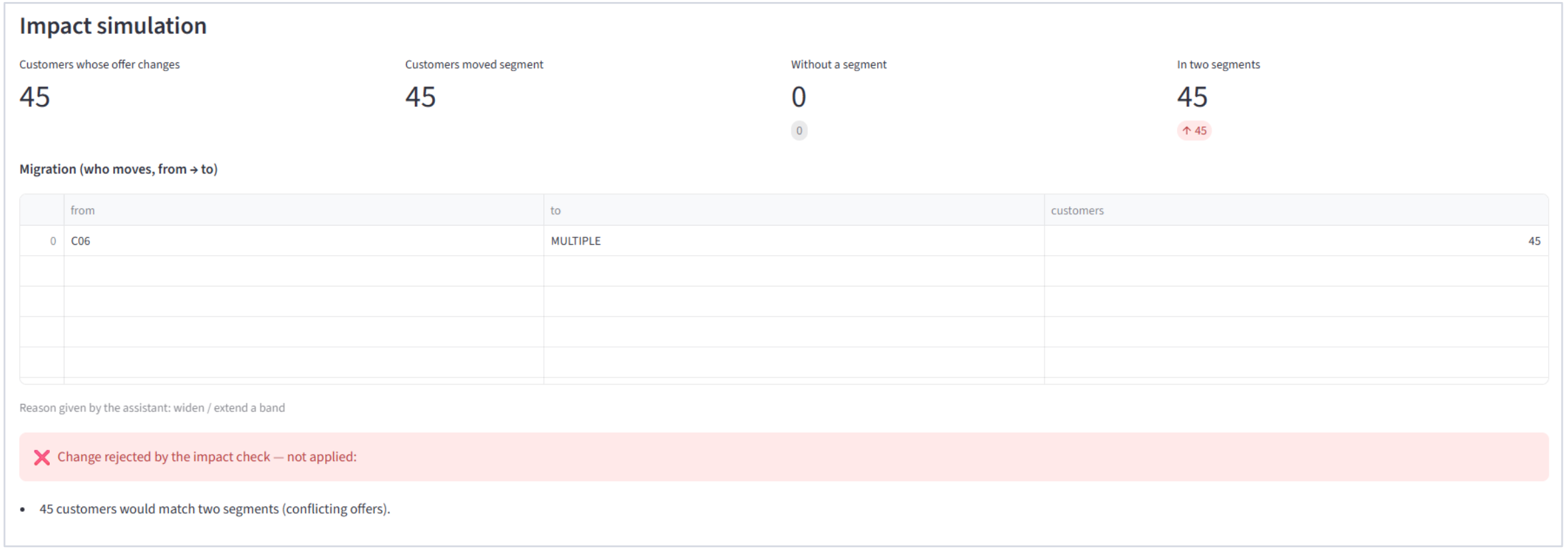
"Extend the Champion Nudge window to 20 days late"



The simulation finds **45 Champions** who would get two conflicting offers: the change is **rejected** and nothing is modified.

LEVEL 2 · APP CAPTURE

The rejection, as the business sees it



Impact simulation

Customers whose offer changes
45

Customers moved segment
45

Without a segment
0
0

In two segments
45
↑ 45

Migration (who moves, from → to)

	from	to	customers
0	C06	MULTIPLE	45

Reason given by the assistant: widen / extend a band

✕ Change rejected by the impact check — not applied:

- 45 customers would match two segments (conflicting offers).

Decision REJECTED_IMPACT · the rules stay unchanged · the decision is logged.

LEVEL 2 · APP CAPTURE

Disabling Silent Last Call: applied, with a warning

Impact simulation

Customers whose offer changes	Customers moved segment	Without a segment	In two segments
221	221	221 ↑ 221	0 0

Migration (who moves, from → to)

	from	to	customers
0	C14	UNASSIGNED	221

Reason given by the assistant: User requested to disable the Silent Last Call segment.

⚠ 221 customers will no longer receive any offer (no active segment covers them).

✅ Change validated, impact-checked and applied automatically. 221 customers affected.

Customers left without a segment = a business choice → applied, with the consequence quantified.

From plain English to a structured change

1 · REQUEST

"make the Regular Nudge segment more aggressive"

2 · RETRIEVE

4 closest rules + 2 variable definitions (Chroma)

3 · GENERATE

gpt-4o-mini proposes a JSON change, or declines

```
{ "segment_id": "C08",
  "field": "offer_intensity",
  "new_value": "VERY_AGGRESSIVE" }
```

This JSON is only a proposal: it then goes through the five barriers of level 2. **The AI proposes, the engine verifies.**

Index once, search on every request

INDEXING · rag/ingest.py

- 19 documents: 14 rules (one sentence each) + chunks of variables.md
- Each document becomes a vector of 1,536 numbers (text-embedding-3-small)
- Stored in Chroma; rebuilt on every indexing run (no duplicates)

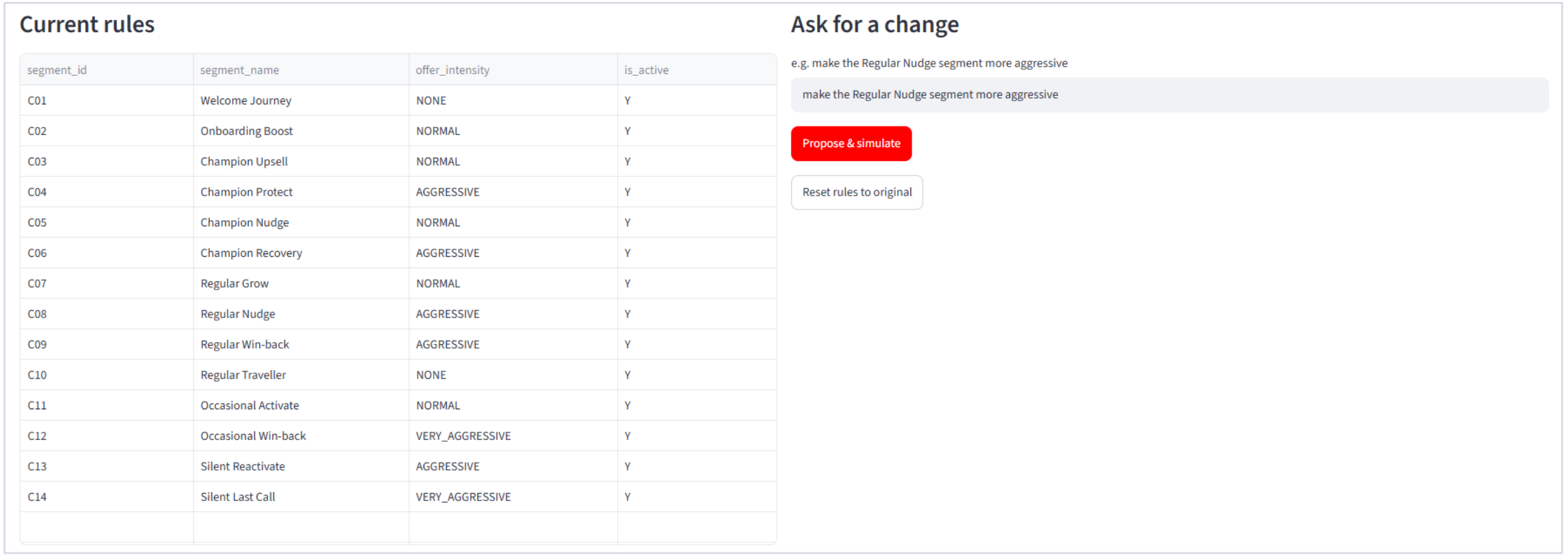
EVERY REQUEST · rag/chain.py

- The sentence is vectorised and compared with the 19 documents
- Context = 4 closest rules + 2 variable definitions
- Prompt = instructions + allowed values + context + grid + request
- gpt-4o-mini, temperature 0, forced JSON output

With 14 rules the whole grid fits in the prompt; vector search pays off when the rule base grows to hundreds of rules.

DEMO · STEP 1 OF 3 · CAPTURE

The business types its request in one sentence



Left: the 14 segments and their current offer. Right: the request, in plain English.

The AI proposes, the engine validates

Proposed change

Change ID: CHG-20261011-134041-969b

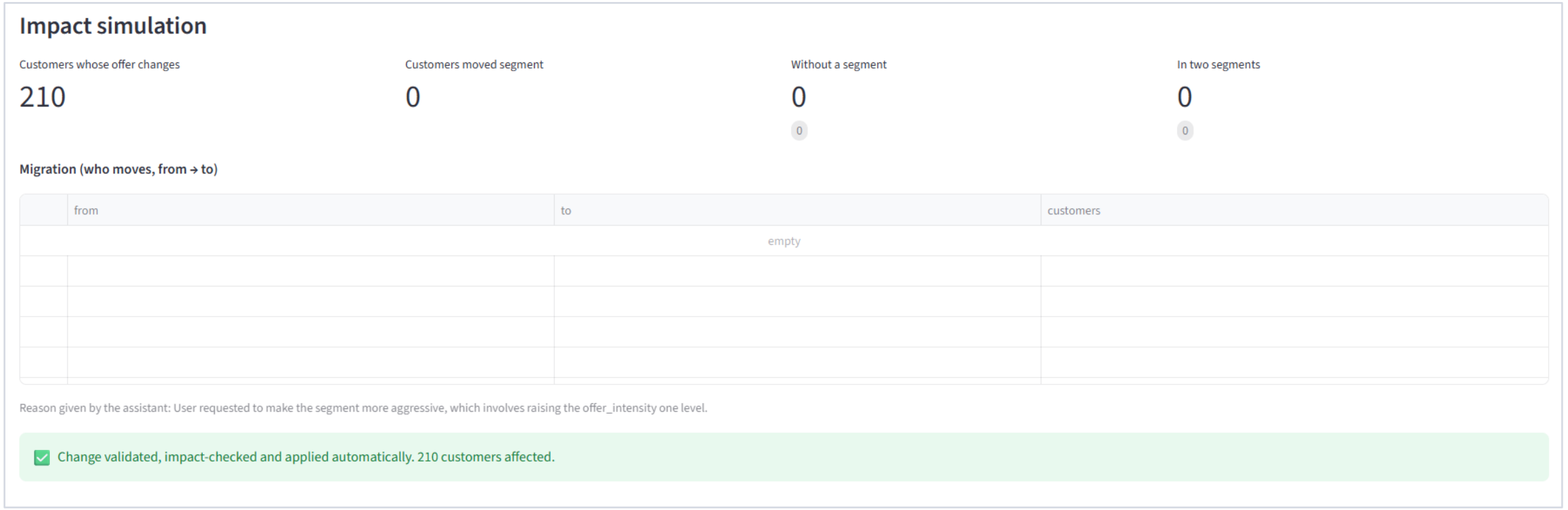

```
{
  "segment_id" : "C08"
  "field" : "offer_intensity"
  "new_value" : "VERY_AGGRESSIVE"
  "reason" : "User requested to make the segment more aggressive, which involves raising the offer_intensity one level."
  "retrieved" : [
    0 : "C08"
    1 : "C05"
    2 : "C09"
    3 : "C11"
    4 : "variables.md"
    5 : "variables.md"
  ]
}
```


Validation passed — C08 · offer_intensity: AGGRESSIVE → VERY_AGGRESSIVE

C08 is retrieved first; the validation shows the value before → after. Each proposal gets a unique change id.

DEMO · STEP 3 OF 3 · CAPTURE

Simulated impact, then automatic application



210 customers get a stronger offer · 0 segment changes · 0 without a segment · 0 in two segments → applied.

5 applied, 6 rejected, each for a real reason

Request (typed in English)	AI proposal	Decision
Make the Regular Nudge segment more aggressive	C08 · AGGRESSIVE → VERY_AGGR.	**Applied · 210**
Be less aggressive with Occasional Win-back	C12 · VERY_AGGR. → AGGRESSIVE	**Applied · 372**
Send a normal offer to Welcome Journey customers	C01 · NONE → NORMAL	**Applied · 411**
Disable the Silent Last Call segment	C14 · is_active → N	**Applied + warning · 221**
Exclude roamers from Champion Recovery	C06 · roaming → N	**Applied + warning · 42**
Extend Champion Nudge to 20 days late	C05 · days-late max 15 → 20	**Rejected · 45 in two**
Make Regular Grow start at 2 renewals	C07 · renewals min 4 → 2	**Rejected · 457 in two**
Target high churn customers in Champion Upsell	C03 · + HIGH risk combos	**Rejected · 294 in two**
Make the Premium Gold segment more aggressive	none: unknown segment	**Declined by the AI**
Add customer age as a targeting variable	none: new variable	**Declined by the AI**
Give everyone a free iPhone	none: out of scope	**Declined by the AI**

Run with the real OpenAI API on the final code. Figures = customers affected.

Four flaws found, four fixes

The AI invented a change

An unknown segment name led to a change on the wrong segment.

Fix: The AI can now answer "no change possible".

Invalid formats

Wrong roaming value, wrong risk format.

Fix: Retrieve variable definitions + list allowed values in the prompt.

An empty change accepted

Once, the LLM proposed a value that was already set.

Fix: "No-op" check: the value must really change.

Double assignments accepted

At first, no impact check: overlaps went through.

Fix: Automatic rejection of any overlap.

Every request leaves a trace

📜 Audit trail

change_id	timestamp_utc	request	segment_id	field	old_value	new_value	decision	reasons	offer_changed
CHG-20261011-134230-cae4	2026-10-11T13:42:30+00:00	reset rules to original					RESET		
CHG-20261011-134227-217f	2026-10-11T13:42:27+00:00	give everyone a free iPhone					REJECTED_NO_CHANGE	No applicable change: The request cannot be expressed as one cha	
CHG-20261011-134222-1ba7	2026-10-11T13:42:22+00:00	reset rules to original					RESET		
CHG-20261011-134218-c5a5	2026-10-11T13:42:18+00:00	add customer age as a new targeting variable					REJECTED_NO_CHANGE	No applicable change: The request asks for a new variable, which c	
CHG-20261011-134212-5bc0	2026-10-11T13:42:12+00:00	reset rules to original					RESET		
CHG-20261011-134209-938d	2026-10-11T13:42:09+00:00	make the Premium Gold segment more aggressive					REJECTED_NO_CHANGE	No applicable change: The segment 'Premium Gold' is not in the cu	
CHG-20261011-134204-36e3	2026-10-11T13:42:04+00:00	reset rules to original					RESET		

Full log: data/audit/change_log.jsonl · rule versions: data/audit/rules_versions/

In the log

Unique id, timestamp, exact sentence, retrieved documents, AI JSON, before → after, decision, reason, impact.

For every applied change

A full copy of the rules before and after: any change can be proven, audited or rolled back.

FINAL STEP · INTEGRATION PLAN

Plugging the app into the Oracle database

1	**Streamlit app**	RAG + guardrails + simulation: only approved changes go out
2	**Oracle staging table**	One row per approved request: who, what, before → after, impact
3	**PL/SQL procedure**	Re-checks, verifies the version, applies in one transaction
4	**Configuration table**	The segment grid, versioned at every change
5	**Daily NBO batch**	Reads the new grid → new offers, with no developer

The app never writes the production grid directly: **the database re-checks every request before applying it.**

Detailed plan: docs/oracle_integration_plan.md · not implemented in the repository, delivered in the target environment.

RESULTS

One offer per customer, every change under control

6,000

subscribers, each in exactly 1 of 14 segments

11 / 11

real LLM requests handled correctly

10

guardrails, each tested with its message

0

overlapping change applied

A strategy change goes from "ticket + developer + release" to **one sentence, controlled and traced in seconds.**

DESIGN CHOICES

Three decisions that make the AI deployable

1

The AI proposes, the engine decides

The LLM never writes the data. Everything goes through deterministic, testable validation.

2

Provider-agnostic

OpenAI today; a private or local model tomorrow, so that no telecom data leaves the company.

3

Safe to show

Synthetic data and a fictional grid inspired by real NBO practice. No operator named.

STACK & SKILLS

From raw data to a business-ready interface

Area	Tools	What it demonstrates
Data engineering	Python, pandas	Variables, rules engine, impact simulation
Governance	Validation + impact check	Safe self-service, no conflicting offers
Generative AI	LangChain, OpenAI gpt-4o-mini	RAG, structured JSON, controlled refusal
Vector search	Chroma, text-embedding-3-small	Indexing a knowledge base
Traceability	JSONL log, CSV rule versions	Audit and rollback
Product	Streamlit	An interface the business can actually use
Quality	Streamlit AppTest, Playwright	End-to-end tests, automated captures
Production (plan)	Oracle, PL/SQL	Staging → checked apply → versioned grid

LIMITS & NEXT STEPS

What I would do next

One change per request

Targeting high churn in Champion Upsell also requires changing Champion Protect.

Next: Grouped, all-or-nothing changes.

Fully automatic application

A change affecting many customers goes through unreviewed.

Next: Human approval above a threshold.

A non-deterministic LLM

Even at temperature 0, a proposal can vary between runs.

Next: A regression test suite run on every release.

A cloud model

Rules are sent to OpenAI: not acceptable with real data.

Next: Private or local model, on-premise.

THANK YOU

The business decides. The AI translates. The engine guarantees.

Live demo on request · code on GitHub

[github.com/your-account/rag-segment-editor]
[Your name] · [your LinkedIn profile]